Rubber Duck Reconstruction Experiments

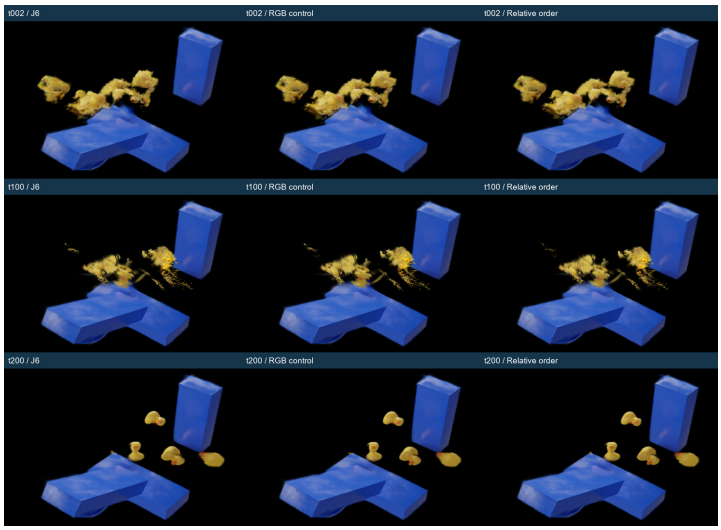
From temporal memory, local structure and springs to pseudo-depth calibration and relative-order training

Report date: 2026-11-06 | Main scope: J6 and L4-L8-R | English edition 1.0

Current conclusion

The relative-order candidate and its RGB control each completed 600 updates. The number of order-consistent relations did not increase, and fragmentation and trailing artifacts did not consistently improve in fixed views. Retain J6; all new weights and negative results are preserved.

Model / trial	All-30 validation PSNR	Measured outcome
J6 parent	31.995 dB	Retained working model; off-trajectory defects remain
RGB continuation, 600 steps	31.931 dB	Matched control
Relative order, 600 steps	31.830 dB	-0.101 dB versus control; visual gates failed



Cover example: camera 188, three times and three models, using identical crops of native renders. See page 15.

Evidence scope: creating this report did not add training or alter frozen results. Relative order and D/alpha are not independent geometric ground truth. The parent model's historical color-conditioned lineage remains disclosed. [S1-S8]

02 / FINDINGS AND READING GUIDE

What this report establishes

Question	Answer supported by the evidence
Did tracking / temporal constraints do nothing?	No. L4 reduced reprojection error for bound memories, but covered only 22 rows and mostly early observations. This did not establish whole-object completeness.
Did connectivity, thickness or springs help?	Local proxies responded: L5 reduced connection gaps and L6 reduced mean strain. Global RGB, area / coverage tails and off-trajectory shape did not improve together.
Can pseudo-depth directly supervise absolute depth?	L7-P and L7-J failed the original scale / dynamic-coverage gates; neither launched Gaussian training. Limited calibration cannot be extrapolated across the image.
Is learning front-to-back order sufficient?	L8-R did train, but fit agreement stayed 9/30 and held agreement stayed 19/32. Whole-duck / environment 3D relations are still not established.
Should training duration or weights simply increase?	The evidence does not support that next step. Broader temporal coverage and reliable relation-to-surface-node correspondence should be checked before preregistering one fixed candidate.

Suggested reading order

Pages 3-4 define the data and comparisons. Pages 5-9 cover temporal memory, structure, springs and scale audits. Pages 10-13 explain relative-order training and numerical results. Pages 14-17 show native-image comparisons; pages 18-20 discuss interpretation, engineering checks and sources.

Three evidence types must remain distinct

1. Training-trajectory RGB tests image fit. 2. Data inference and parent-model proxies test consistency of assumptions. 3. Independent geometric GT tests actual 3D correctness. These Rubber Duck experiments do not provide the third type for fixed-time, multiview dynamic geometry.

"Not promoted" means an experiment completed without replacing the retained working model. It does not mean the model was never updated or its new weights were not saved. [S1-S8]

03 / SHARED EXPERIMENT CONDITIONS

Data, cameras and evidence

Item	Verified meaning
Target dataset	The PIDG Rubber Duck benchmark. No new physical-camera capture was performed for this work.
Training observations	120 RGB images, 120 distinct exact times and 120 camera transforms. Simultaneous multiview observations: 0.
Ordinary RGB training	Native 1600 x 900; 0.8 L1 + 0.2 DSSIM. Original RGBA is composited on black; alpha is not introduced as a new label.
Validation	All 30 validation images at native 1600 resolution. This is separate from the auxiliary loss's held set.
Fixed views	Camera IDs 86 / 188; time numbers 2 / 100 / 200, with normalized t = (number - 1) / 199. IDs are not training-list indices.
Unified dynamic pool	15507 dynamic Gaussian rows and a static library of 4126 points. These are model capacity, not a duck count.
Known and unknown	Insufficient visibility, binding or boundary evidence remains unknown. Darkness or residuals do not justify labeling background / dynamic or deleting points.

Seeing every surface over time does not automatically determine a full 3D model

Camera changes and object motion are coupled across time. Fusion requires a persistent correspondence between the surface seen now and the surface seen earlier, followed by correct motion compensation. Similar 2D appearance alone does not provide simultaneous multiview depth intersection or reliable front / back surface pairing.

J6 displacement and D/alpha can support hypotheses, but cannot independently prove those same hypotheses. Lower pixel error can coexist with flattened sheets, strands and dark yet opaque support.

Training metadata and the validation list were checked read-only during report generation. Test data were not used to select training points. [S1, S8, S15]

04 / MODEL AND EXPERIMENT FAMILY

Lineage and fair comparisons

What is J6?

J6 combines I10 dynamic state with the user-accepted J4 static library, then compares frozen static parameters against trainable static opacity / SH for 2000 steps per arm. Joint J6 reached 31.994673 dB validation PSNR, +0.593549 dB over its matched control. The RGB gain was retained, while cleanup / completeness gates still failed. The combination start was 29.068226 dB and the frozen-static control reached 31.401124 dB; joint SSIM was 0.980425.

Experiment	Steps / arm	Candidate adds	Control
L4 Memory	2000	Measured cross-window UV memory and two-frame projected displacement	RGB
L5 Support	2000	Connectivity, local support thickness / area and observed-footprint contribution	RGB + same L4 mechanism
L6 Springs	2000	Local length strain with shared fixed k = 1	RGB + same L4 mechanism
L7-P / L7-J	0	Pseudo-depth qualification and joint scale audits	Read-only inference; no training arms
L8 Intervals	0	Calibrated absolute-depth interval separation	No fit pairs; no training
L8-R Order	600	Temporally stable order from raw relative inverse depth	RGB

L4 / L5 / L6 / L8-R each start from the same complete J6, including two Adam states, RNG and remaining camera stack. Budgets and control losses differ; L5 and L6 controls also follow different CUDA numerical trajectories. Compare each matched increment, rather than ranking all candidates as equal-budget repeated seeds.

These are separate branches from J6, not sequential L4-to-L5-to-L6-to-L8-R continuation. The historical 40-step CUDA trajectory failure remains recorded. Single matched trials have no repeated-seed confidence intervals. [S1-S8]

05 / MEASURED CROSS-FRAME SURFACE MEMORY

L4: what temporal memory constrained

Method: real observation memory, not synthesized unseen views

730 bidirectional RGB tracklets formed 641 memory units; 89 fit overlap links produced 75 long memories. Requirements included mutual-nearest matching, overlap RMS at most 0.75 px, maximum at most 1.5 px and ambiguity margin at least 0.25 px. Odd held frames only checked selected links. Ultimately 22 passed canonical binding and native own-row visibility, spanning 16-20 frames.

The candidate retrieves up to 4 real fit keyframes. Current projection, keyframe projection and two-frame projected displacement of shared canonical nodes receive gradients, with weight 0.01. Known per-frame cameras prevent treating raw image displacement as world motion. Unobserved frames and occlusion gaps receive no invented targets.

Evidence	Measurement	Interpretation
Actual coverage	22 rows, 372 observations; time thirds 356 / 16 / 0	Mostly early local memory, not whole-duck surface area
Additional times	18 memories add 76 measured observations outside the root window	More information, without proof of newly complete surfaces
Held reprojection	J6 0.860543; control 0.895909; candidate 0.561905 px	Bound local correspondence improves; held still receives RGB training
Actual loss usage	216 / 2000 steps; 4734 keyframe-pair applications	Not every step or every surface receives supervision
All-30 val PSNR	Control 31.636682; candidate 31.319321 dB	-0.317361 versus control; -0.675352 versus J6; both gates fail

Established and unestablished

Gradients and improved reprojection confirm an effect on bound local tracks. They do not establish whole-object connectivity, thickness or off-trajectory shape. Fixed views 86 / 188 still show fragments and high-alpha trails.

Qualification refreshes every 250 steps: unknown suspends, 2 contradictions revoke, 2 passes restore. No new Transformer / RNN, point growth or unseen-surface triangulation was introduced. [S2]

06 / LOCAL STRUCTURE CONSTRAINTS

L5: connectivity and surface support

Both 2000-step arms start from J6 and share L4 temporal memory. The candidate adds four averaged structural terms with total weight 0.01 and at most 3 qualified support groups per step. This is neither object segmentation nor true body-thickness supervision.

Component	Implementation	Limitation
Connection / collapse	2-sigma ellipsoid gap along the center line; half-parent-length lower bound	A parent reference can preserve old incorrect support
Support thickness	Smallest principal-axis width of covariance plus center distribution, held within 0.8-1.25 times parent	Not physical front-to-back thickness or wall thickness
Tangent area / scale	Product of two largest axes in the same range; maximum point scale at most 1.25 times	No guarantee of closure or missing-surface generation
Observed footprint	Native depth-sorted group contribution at least 0.8 times parent	Own visible-support proxy, not independent GT

Fixed 360 patch-frame metric	Temporal control	L5 candidate
Mean connection penalty	0.044638	0.018335 (-58.92%)
Thickness outlier patch-frames	2	3
Area outlier patch-frames	19	13
Contribution below 0.8 times parent	96	67
All-30 val PSNR	31.223379	31.859706 (+0.636327 dB)

Initial support: 60 patches, 275 distinct dynamic rows and 337 within-patch edge applications; start thirds 59 / 1 / 0. The 1.77% row coverage is not surface area. Structure was used on 166 steps, with 1174 patch and 5527 edge applications. Middle / late times and unrepresented surfaces remain poorly covered.

The candidate is -0.134967 dB versus J6 and passes both RGB non-degradation gates, but fixed-view fragmentation and dark high-alpha tails do not consistently disappear. A smaller local gap does not prove better complete geometry. Parent contribution exists for 359 / 360 patch-frames; the remaining unknown is not counted as zero error. [S3]

07 / SHARED FIXED-STIFFNESS GEOMETRIC REGULARIZATION

L6: benefits and costs of springs

The trial again starts from J6 with a matched temporal control; it does not stack L5 thickness / area / footprint losses. Each reliable edge uses the median parent length over 6 fit frames as a fixed reference, shared k = 1 and total weight 0.01. No mass, contact, external force or material identification is added.

strain = (l(t) - l_ref) / l_ref
$E = \mathrm{sqrt}(1 + \mathrm{strain}^2) - 1$

Proxy over the fixed initial denominator	Control	Spring candidate
Mean absolute strain	0.108432	0.092034 (-15.12%)
Mean spring energy	0.011789	0.008677 (-26.40%)
Most negative edge strain	-0.411848	-0.695519
Area outlier patch-frames	9	16
Contribution below 0.8 times parent	67	76
Validation PSNR	31.938599	31.589028 (-0.349571 dB)

Why lower mean strain did not yield a complete shape

Average length regularization mainly reduces local stretch. It creates no surfaces and does not ensure that area or compression tails improve. Connections still cover only 60 patches / 275 rows, with start thirds 59 / 1 / 0. The parent median is not a physical rest length and may suppress genuine flexible deformation.

Springs were used on 178 / 2000 steps, with 1439 patch and 6608 edge applications. PSNR drops 0.405645 dB versus J6; both 0.3 dB control / parent gates fail. Strands, fragments and dark support persist, so the candidate was not promoted.

What this says about spring-mass constraints

The experiment tests a local spring geometry regularizer, not full mass-spring dynamics. Its result does not support increasing stiffness or adding mass / unknown forces to compensate for correspondence and coverage gaps. [S4]

08 / SCALE, MOTION COMPENSATION AND BOUNDARIES / NO GAUSSIAN UPDATES

L7-P: pseudo-depth qualification

The fixed official Depth Anything V2 Small processes all 120 training RGB images at observation resolution 400 x 225, with 518 preprocessing and FP32 evaluation. It is neither changed nor fine-tuned. Finite predictions on black background are not automatically surfaces.

Scale and motion checks

Triangulated anchors from stationary-compatible RGB tracks fit a positive-affine inverse-depth model per frame. Spatial fit / check separation and two-fit-subset stability must pass; only interpolation within the fit convex hull is allowed. Parent J6 rendered depth is not scale-fitting GT.

$z(u) = 1 / (a\, r(u) + b),\ a > 0$
$X_hat(t) = T_camera(t)^{-1}\ [z\, K^{-1}\ p] - displacement_J6(t)$

Camera backprojection first removes the change of camera coordinates; J6 time-varying displacement is then subtracted to check a track's world consistency. This is parent-conditioned compensation, not proof of binding or true motion. Even frames fit, odd frames check; each side requires at least 4 valid observations.

Qualification measure	Observed	Original gate
Calibrated views	24 / 120	At least 30
Early / middle / late	7 / 7 / 10	At least 8 per third
Continuous-motion checks passed	0 tracks	At least 30; at least 5 per third
Distinct dynamic rows	0	At least 25
Native-qualified track observations	45 / 1464 locally usable	At least 4 fit and 4 check per track

There are 122 native-qualified tracks x 12 observations = 1464. At most 7 / 12 depths are valid for any track, so no track reaches the numerical motion-compensation test. This does not show tracking or compensation to be ineffective. Of 53196 boundary observations, 42276 remain unknown. The limited known subset does not show higher boundary failure and cannot establish occlusion as the sole cause. [S5]

09 / DATA-INFERENCE AUDIT / NO GAUSSIAN UPDATES

L7-J: joint scale and more anchors

1981 new full-image texture seeds yield 963 RGB-qualified tracks and 265 stationary-compatible tracks; adding 362 old tracks gives 627. Cross-window duplicates form 507 groups, split as whole groups into fit / check and two fit subsets. Per-frame 4 px deduplication leaves 5351 anchor observations. Static points are not selected by blue color.

120 normalized affine inverse-depth models are fitted jointly with fixed Huber error, a 0.05 second-time-difference prior, positive-slope bounds and tiny ridge. Smoothness is a prior, not evidence of correct scale; held data can only check, not adjust coefficients.

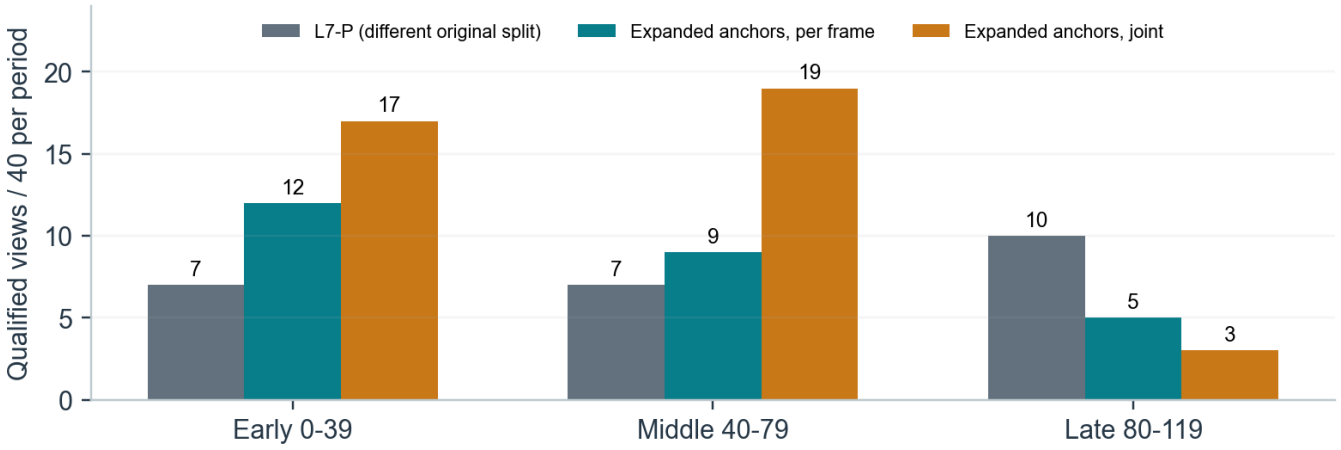


Figure 1. Each third contains 40 frames. Joint and per-frame fits with the same expanded anchors are comparable. Original L7-P uses a different split and is not a strict single-factor control. [S5-S6]

Metric	Same anchors, per frame	Joint across time
Qualified views / 120	26	39
Locally usable dynamic observations / 1464	16	75
Tracks reaching the motion test / 122	0	2
Motion checks passed	0	1 (5 fit depth proposals)

Only track 052_020, dynamic row 6886, passes: 11 local observations and held compensation residual median 1.842%, P90 3.623%. These are not independent GT. Only 3 / 40 late views qualify, and dynamic track / row gates fail; the 5 depth proposals are not trained. Potential fit-convex-hull image area rises 6.846% to 12.884%, not accurate surface coverage. [S6]

10 / INTERVAL FAILURE PRESERVED / SEPARATE SCALE-FREE CANDIDATE

L8 and L8-R test different hypotheses

Design	L8 Interval order	L8-R Raw temporal order
Order source	Separation of full / split calibrated absolute-depth intervals	Small raw relative inverse-depth patch intervals
Metric scale	Requires stable calibrated intervals	No absolute-scale fitting
Dynamic tracks	L7-J tracks passing metric checks	Original 122 T4 native-qualified tracks
Cross-frame checks	Interval reliability and spatial support	At least 4 even same-sign and 4 odd checks
Training / diagnostic pairs	Fit 0 / 94; held 1 / 43	Fit 30; held 32; all hypotheses retained
Actual training	0 steps	600 steps per arm

Why L8 did not train

Among 94 fit proposals, 77 have overlapping absolute intervals and 17 have weak raw order, leaving 0 training pairs. The single held pair remains diagnostic and is not moved into training. This outcome retains the original gates; permission to try training does not manufacture valid targets.

How L8-R separately tests relative front-to-back relations

A scale-free single candidate was separately preregistered. It skips metric intervals but requires raw order to persist through the observation window, remain locally smooth, and agree across fit / check times. It neither rewrites the failed L8 result nor relaxes old global geometry admission gates. The authorization is for a local exploratory trial, not automatic inference of whole-duck relations.

Identity and order remain separate

Endpoints are a measured UV bound to a dynamic row and a stationary-compatible RGB-anchor UV. They carry no duck ID or environment semantics. A maximum of 4 pairs per image is not an object count. [S7-S8]

11 / CORE MECHANISM OF THE COMPLETED TRAINING TRIAL

L8-R: targets and the loss

Let S be image q90 - q10 of raw relative inverse depth. A strong sign requires both 5 x 5 endpoint patches to have range at most 0.1 S, endpoint distance at least 8 px, and interval separation at least 0.1 S. Larger inverse depth means nearer. Missing, boundary or contradictory evidence remains unknown.

Within 12 frames of a dynamic-track / anchor-group hypothesis, at least 4 even fit frames must support one sign s, and at least 4 odd check frames must confirm it. Check can reject but cannot choose a replacement sign. Whole anchor groups are separated; stable track / group ordering caps each image at 4 pairs, with fixed confidence 0.5.

Full construction denominator	Fit library	Diagnostic held library
Group hypotheses	4540	2953
Temporal support passed	41	27
Frame proposals before cap	193	129
Final fixed relation pairs	30	32
Dynamic rows	6498, 15419	6498, 15419
Frame indices	8 even frames in 100-114	8 odd frames in 101-115

At native 400 joint-render resolution, bilinearly sample D and alpha, then set z = D / alpha. Require positive finite depth, alpha > 0.5 and current dynamic-row reprojection within 4 px. The signed log-depth gap is:

$$g = s\,[\log(z_{anchor}) - \log(z_{dynamic})]$$

$$e = \max(0, \log(1.02) - g)$$

$$L_{order} = (0.001 / N) \sum [0.5 * valid * (\sqrt{1 + (e / 0.05)^2} - 1)]$$

N is the original fixed per-frame proposal denominator, not the shrinking valid subset. Target sign, confidence and qualification are detached. Gradients pass only through current native D/alpha to dynamic canonical / grid / deform parameters. s = +1 means the dynamic endpoint should be nearer. No absolute depth or large depth gap is specified. [S8-S9]

12 / ACTUAL TRAINING AND QUALIFICATION HISTORY

Revocable eligibility, sparse usage

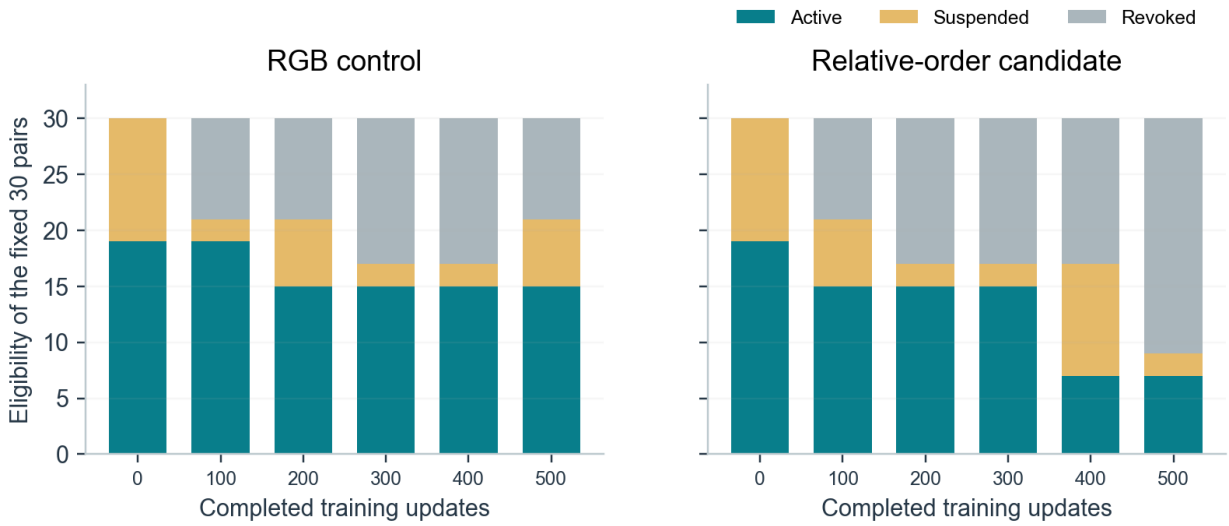


Figure 2. The fixed 30 pairs refresh every 100 steps. Initially 19 active / 11 suspended; after the candidate's step-500 refresh: 7 active / 2 suspended / 21 revoked. Final eligibility carries over from that refresh; there is no extra step-600 check. [S10]

Check / training item	Fixed rule or measured result
Own contribution and RGB	Native single full-scene depth sorting; own-row patch contribution share at least 0.01. At dynamic UV, observed / rendered patch NCC at least 0.6 and both standard deviations at least 2 on a 255 scale.
Revoke / restore	Unknown immediately suspends; 2 contradictions revoke; 2 passes restore. Current projection, alpha and positive depth are also checked every step.
Training budget	600 native-1600 RGB updates per arm. Only the candidate adds a 400-resolution order render. Fixed Adam epsilon 1e-15.
Actual order usage	20 / 600 nonzero auxiliary steps; 74 valid pair-observation applications at indices 102 / 104 / 106 / 112.
Frozen state	Static model / Adam, physics / Adam / graph caches and topology. No L4 / L5 / L6 losses are stacked.

A 1% own-contribution threshold is only a visibility gate, not evidence that the row dominates joint depth. Other Gaussians along the ray can substantially affect D/alpha. This limits the link between image order and a specific surface-node relation.

20 / 600 is about 3.33%. The 600 RGB updates must not be described as 600 depth-constraint updates. Losses and gates stayed fixed; weights and duration were not automatically increased after results. [S8-S10]

13 / MATCHED NUMERICAL RESULTS AND CROSS-TRIAL CONTEXT

Results: order counts did not improve

Model	PSNR dB	SSIM	L1
J6 parent	31.994673	0.980425	0.00329669
Current RGB control	31.930919	0.980417	0.00331917
Current relative order	31.829998	0.980195	0.00336260

Order diagnostic (not GT)	J6	RGB control	Candidate
Fit agreement / fixed 30	9/30	9/30	9/30
Fit valid / fixed 30	28/30	28/30	28/30
Held agreement / fixed 32	19/32	19/32	19/32
Held valid / fixed 32	32/32	32/32	32/32

Candidate PSNR is -0.100921 dB versus control and -0.164675 dB versus J6. Both maximum-drop 0.3 dB gates pass, which establishes only limited non-degradation. Mean fit signed log-gap changes -0.026816 to -0.024436; held changes 0.022797 to 0.024123. No pair changes its prior-order side or validity.

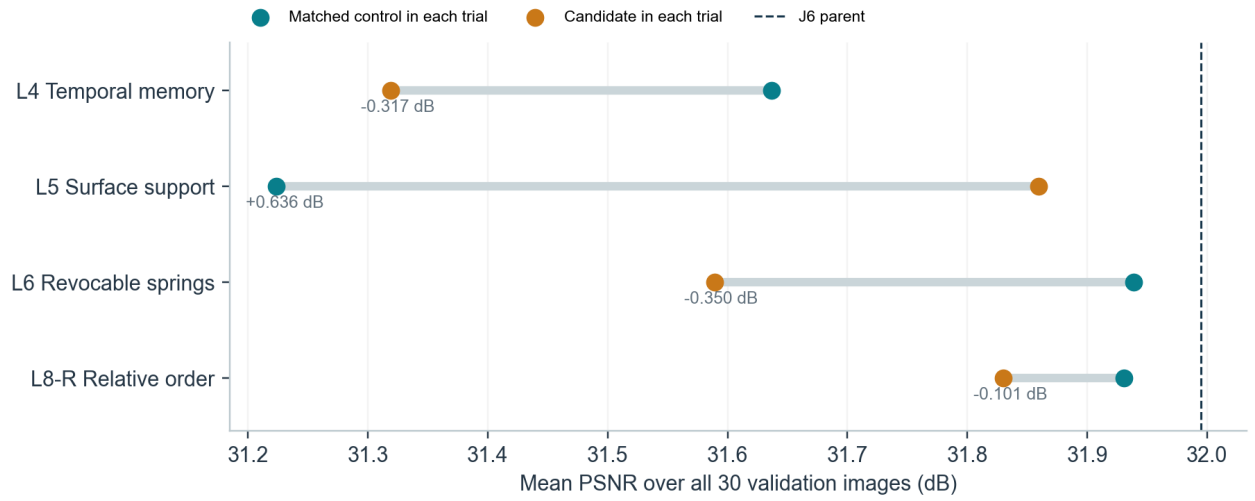


Figure 3. Each candidate is compared with its own matched control. L4 / L5 / L6 use 2000 steps; L8-R uses 600. These are neither repeated seeds nor a ranking against one shared control. [S2-S4, S8]

Across all 122 tracks / 1464 observations, held reprojection is 0.911032 px for the candidate, 0.909937 for control and 0.888095 for J6. Binding does not improve globally. Held excludes auxiliary targets but still receives RGB training; it is not independent geometric generalization. [S8]

14 / IDENTICAL CROPS OF NATIVE 1600-RESOLUTION RGB

Camera 86: early, middle and late

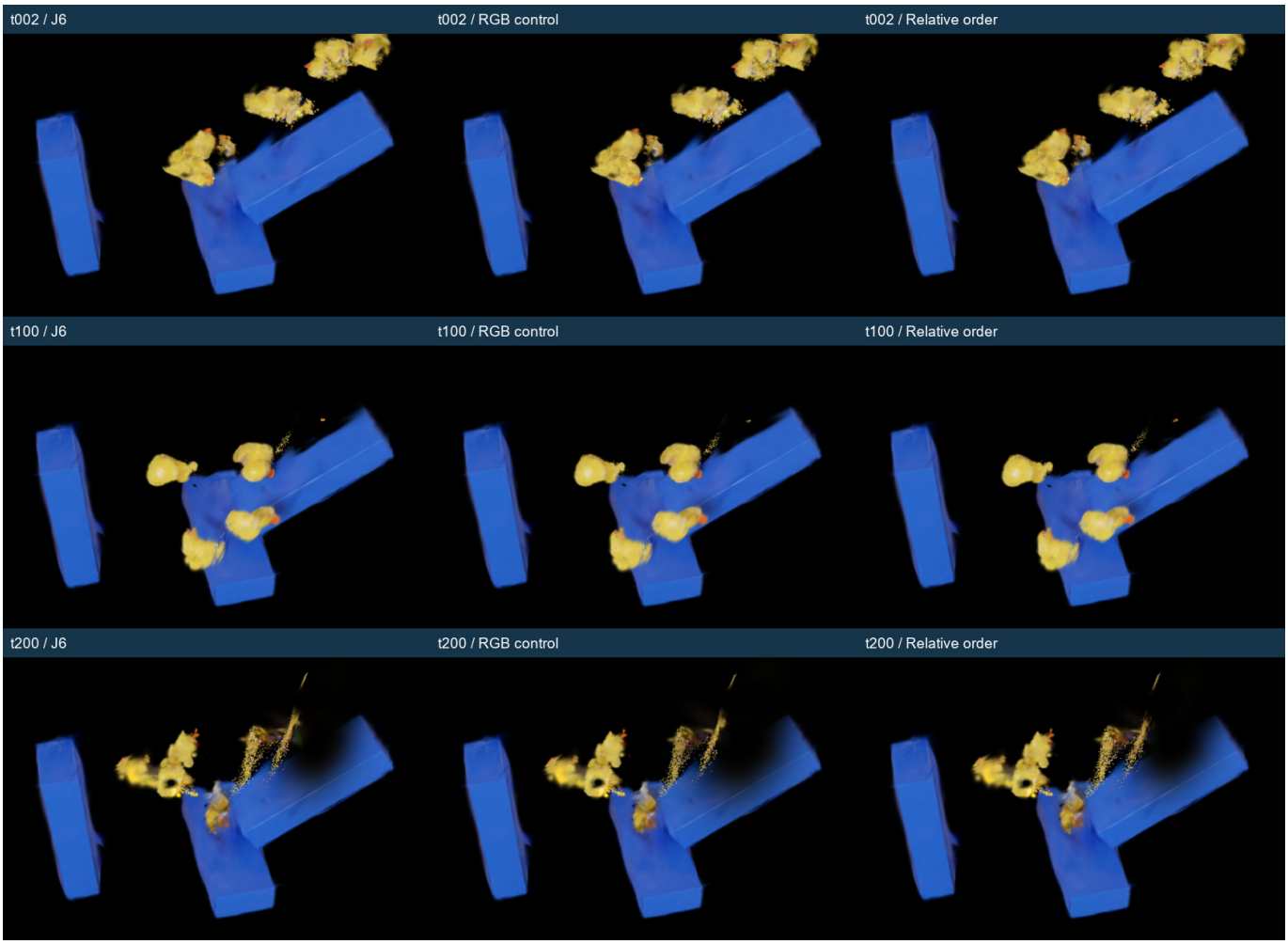


Figure 4. Columns: J6 / current RGB control / relative-order candidate. Rows: time numbers 2 / 100 / 200. Identical native-RGB crops, without repainting or retouching. [S11]

Observations

Early surfaces remain loose with visible gaps. A recognizable middle-time outline does not remove posterior particle chains. All three late-time models retain stretched particle sheets, fragmented regions and dark support. The candidate does not consistently join fragments into complete objects.

The static layer is fully frozen, so the blue support structure is unchanged. Different dynamic occlusion is not a static reconstruction improvement. There is no independent RGB / geometry GT for these fixed-camera cross-time combinations; the images support defect comparison, not calibrated geometric error. [S11]

15 / IDENTICAL CROPS OF NATIVE 1600-RESOLUTION RGB

Camera 188: strands and gaps persist

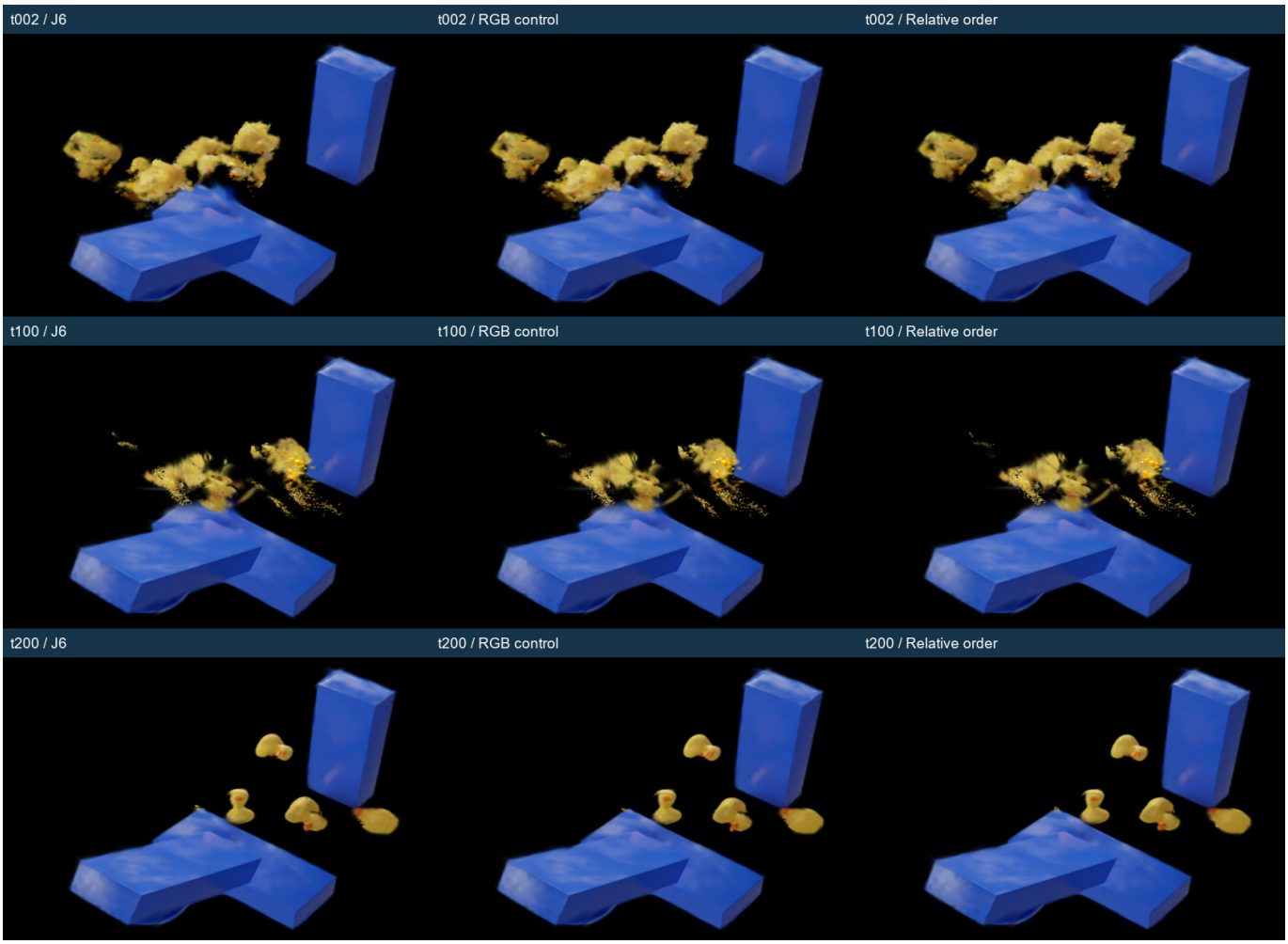


Figure 5. Columns: J6 / current RGB control / relative-order candidate. Rows: time numbers 2 / 100 / 200. Sampling matches Figure 4; each camera uses its own fixed crop. [S11]

Observations

Early duck bodies remain fragmented. Middle-time images show laterally extended particle bands and strands. Late RGB looks more recognizably duck-like, but that does not establish removal of dark high-alpha structures. Differences are mainly local details, without consistent completion.

Unchanged local order and broadly similar renders agree in this trial. Neither observation disproves all relative priors or RGB solvability. Current supervision touches only 2 dynamic Gaussian rows at late times. [S8, S11]

16 / JOINT REVIEW OF DYNAMIC RGB, ALPHA AND D/ALPHA

Darker RGB is not removed geometry

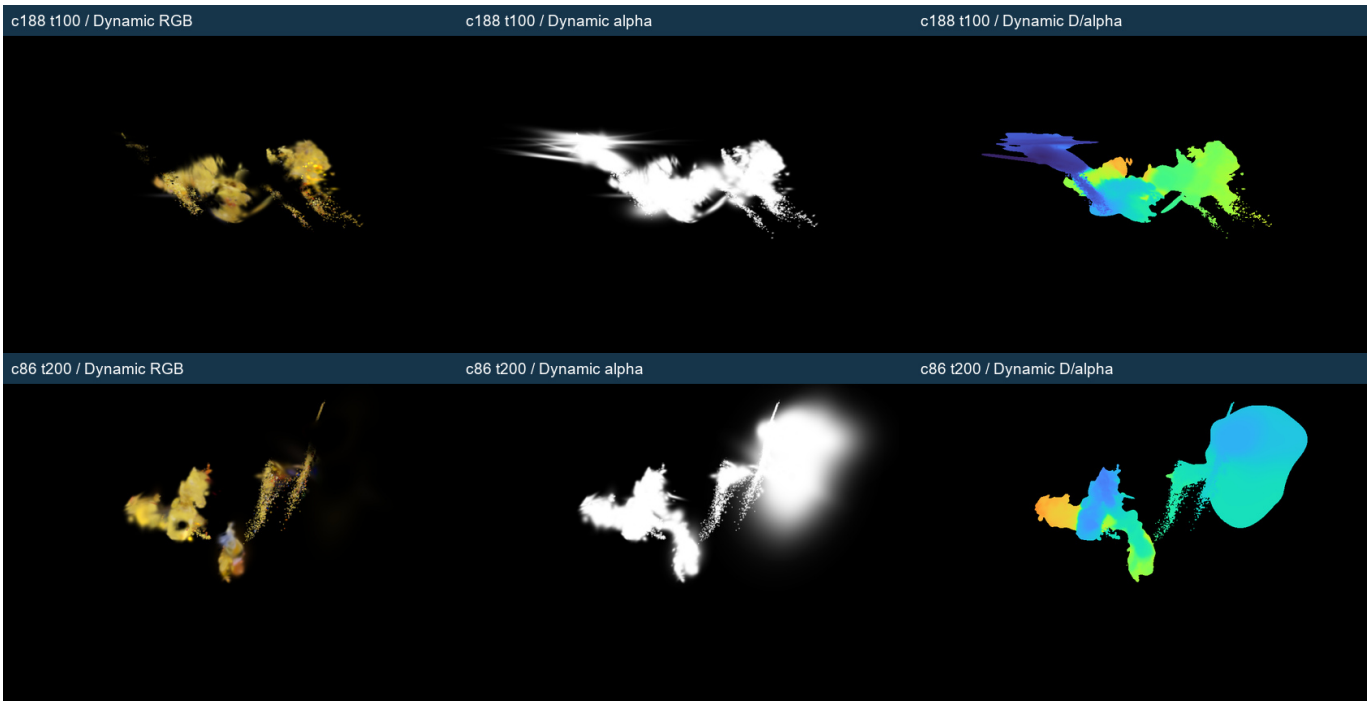


Figure 6. Relative-order candidate: c188 / t100 above, c86 / t200 below. Columns: dynamic RGB / dynamic alpha / dynamic D/alpha. Depth colors reuse the same parent-static range within each camera. Black low-alpha regions are missing, not zero error. [S11]

Dark lateral bands in RGB still show clear alpha and D/alpha support. A darker color does not establish deletion of incorrect geometry. Conversely, high alpha plus dark color does not justify classifying background or deleting points.

Dark RGB and alpha > 0.5, six fixed images	Pixel count	Comparison
J6	107498	Parent reference
Current RGB control	121627	Above J6
Relative-order candidate	113170	-6.95% versus control; +5.28% versus J6

Defined only for evaluation: every dynamic RGB channel < 0.05 and alpha > 0.5. Denominator: 1440000 pixels per image, 8640000 pixel applications total. This is not independent-GT area or color-based training classification. Both visual cleanup and completeness gates fail. [S8, S11]

17 / DIFFERENT MECHANISMS, PERSISTENT DEFECTS

Earlier branches in the same view

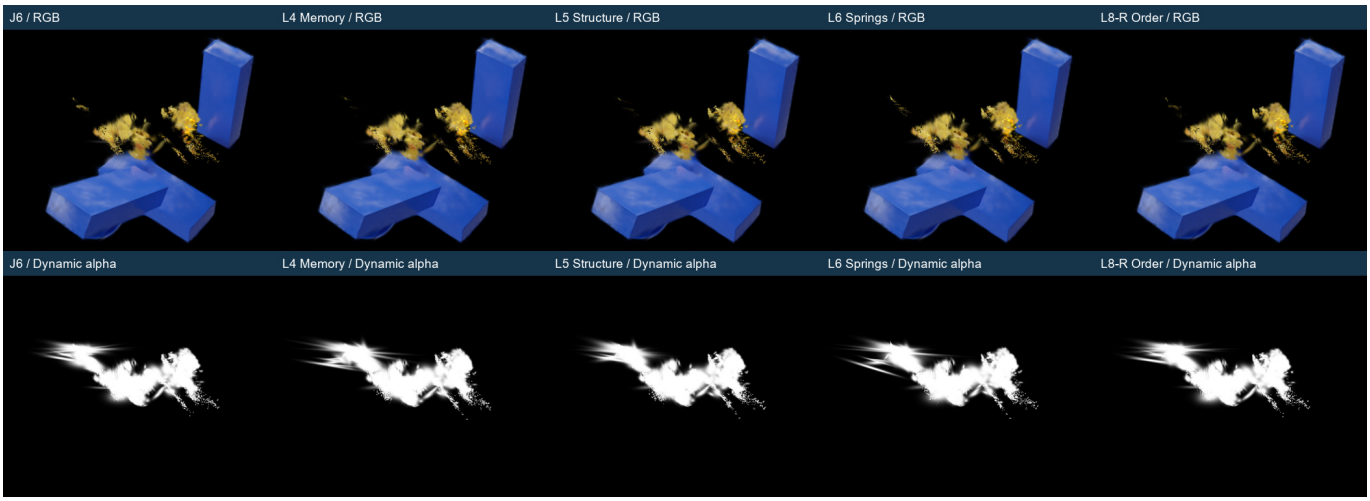


Figure 7. Same world region at camera 188 / t100. Columns: J6, L4, L5, L6, L8-R. RGB above, dynamic alpha below. Different budgets and losses prevent a strict equal-budget ranking. [S2-S4, S8, S11]

Shared observations across trials

Temporal memory, local structure, springs and relative depth change local details, yet fragmentation and lateral high-alpha support remain. Their proxies differ: pixel reprojection, parent support geometry, parent median edge length and monocular-prior order. Self-consistency of any one proxy does not validate an entire duck surface.

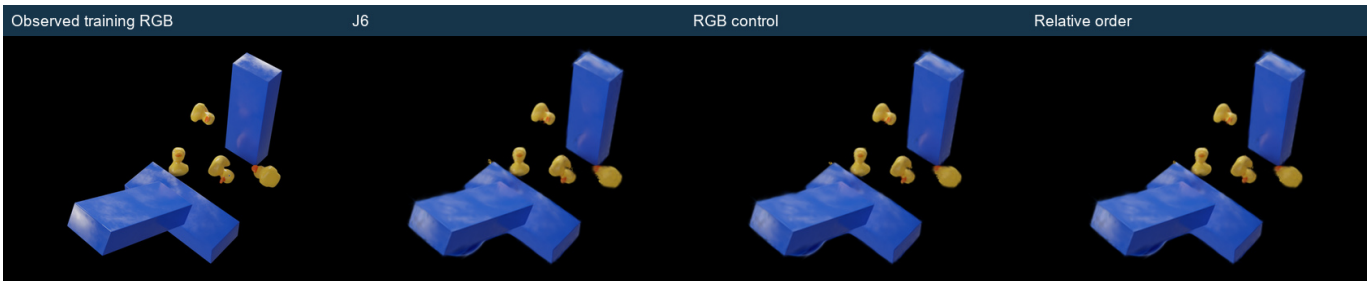


Figure 8. Observed training image at list index 119 and the three original-trajectory renders, rearranged from existing images. Appearance is cleaner here. This is a training camera / time observation, not independent GT for fixed cameras 86 / 188 across time. [S11]

Figure 7 shows candidates, not their matched controls; matched numerical increments are on page 13. Figures 7 and 8 have different time / camera conditions and illustrate evaluation scope. Subtracting them would not measure geometric error.

18 / TECHNICAL LIMITATIONS AND ACTIONABLE CHECKS

What is missing, and what to test next

Supported limitation	Do not overinterpret	Priority check
Sparse temporal support: L4 / L5 / L6 are early-heavy; L8-R has only 2 late rows	This does not show that no other usable surfaces exist in the RGB sequence	Use fixed denominators by time, view and visible surface; expand reliable binding before training
Uncertain 2D-track to canonical-node / patch correspondence	Row counts and graph links are not object identities	Keep revocable links; independently check correspondence across occlusion and surface membership
Joint D/alpha mixes Gaussian contributions; own-share floor is only 1%	Order gradients need not modify the intended surface node	Report contribution dominance and surface association before designing a relation loss
Parent reference length / thickness / depth may preserve wrong support	Better self-proxies do not prove real geometry	Add mutually exclusive observation checks; protect only evidenced surface relations
No surface growth or reliable front / back pairing	More regularization cannot create missing surfaces automatically	Require testable evidence for new surfaces and thickness before imposing either

Recommended sequence, not an executed result

First audit surface nodes and visible contributions over broader times / views, retaining unknowns and original candidates. If evidence is sufficient, fix one node-level relation or local surface-fusion mechanism; declare the matched budget, thresholds, native-gradient and recovery checks, then run one trial. Stacking several losses at once would obscure attribution.

Current decision

Retain J6. Preserve the complete L8-R weights without promotion. Do not automatically extend training, raise weights, sweep parameters or alter gates. This negative result does not invalidate temporal information, all pseudo-depth, or RGB solvability. Physical mass, constitutive laws and forces were not added.

These limits follow from fixed denominators, native images and implementation behavior. Validation / test data did not select new training points. [S1-S11]

19 / CORRECT EXECUTION AND GEOMETRIC SUCCESS ARE DISTINCT

Engineering checks and retained failures

Verification	Completed in this trial
Tests / default off	20 relevant tests. Native RGB / alpha / D exact equality and original gradient tolerances pass.
Recovery contract	User-accepted exact restoration of full model, two Adam states and bindings, RNG, camera stack and physics graph. Adam updates are exact with the same native gradients.
Gradients	Actual order loss is computable. Qualification-only forced violations verify finite nonzero canonical / grid / deform gradients; they are not training targets.
Full replay	1200 camera samples; exact native replay of 10 saved refresh models from steps 100-500; independent eligibility-transition checks.
Artifacts	180 image SHA / readability checks; all fields of 6 full-Gaussian PLYs plus 9 world-XYZ diagnostics; 2 H264 videos fully decoded, 200 frames each.
Freeze / closure	608 frozen-file SHA checks and 17 task-frozen Git raw-byte checks. pending:false; task lock released.

First 100-step failure and independent repair

The first formal RGB control exited after 100 completed updates because stale measurement metadata collided with a new refresh field. No complete step-100 checkpoint existed, so partial state was not resumed. Failure stdout, initial state and progress are retained. Independent repair_run strips only old measurement metadata, then calls the frozen original transition logic.

After more than 600 supplemental state calls and revoke / restore checks, both arms again passed short training and independent recovery. Each then restarted from the original complete J6 for 600 steps. The final comparison contains 1200 updates, separate from the failed control's 100 and two qualification rounds. Original epsilon 1e-15, losses, gates and historical 40-step CUDA failure remain unchanged.

Commits: 5e1a0800 preregistration; aa62478d first qualification; 6b2d6f3c repair record; 4a8173d8 requalification; eb70f4e4 training; 53b7fbfc results; 9874a28d final validation. Browser playback was not tested. Manual review used original images and decoded contact sheets, not manual frame-by-frame playback. [S8-S14]

20 / REPOSITORY EVIDENCE INDEX

Sources, weights and review entry points

Paths below are relative to F:/A_Edinburgh/PhysGauss. The companion manifest records SHA hashes for report sources and every source image. This English edition preserves the Chinese original and all historical experiment files.

[S1] J6 method, lineage and results experiments/staged_reconstruction/RubberDuck_I10_joint/v1/assessment.md

[S2] L4 temporal method and results experiments/staged_reconstruction/RubberDuck_temporal_memory/v1/assessment.md

[S3] L5 structure method and results experiments/staged_reconstruction/RubberDuck_surface_support/v1/assessment.md

[S4] L6 spring method and results experiments/staged_reconstruction/RubberDuck_revocable_springs/v1/assessment.md

[S5] L7-P pseudo-depth qualification experiments/staged_reconstruction/RubberDuck_pseudo_depth/v1/assessment.md

[S6] L7-J joint scale experiments/staged_reconstruction/RubberDuck_joint_scale/v1/assessment.md

[S7] L8 zero-fit interval failure experiments/staged_reconstruction/RubberDuck_ordinal_depth/v1/assessment.md

[S8] L8-R report / protocol / evaluation
experiments/staged_reconstruction/RubberDuck_relative_order/{protocol.json,v1/assessment.md,v1/evaluation.json}

[S9] Order implementation variants/staged_reconstruction/{i10_relative_order,i10_ordinal_depth}/core.py

[S10] Usage and refresh audit experiments/staged_reconstruction/RubberDuck_relative_order/v1/{controller_audit.json,pairs_audit.json}

[S11] Native images, PLYs and videos
output/staged_reconstruction/RubberDuck_{relative_order,temporal_memory,surface_support,revocable_springs}/v1/

[S12] Independent artifact audit experiments/staged_reconstruction/RubberDuck_relative_order/v1/independent_audit.json

[S13] Recovery and native gradients experiments/staged_reconstruction/RubberDuck_relative_order/v1/{recovery.json,native_check.json}

[S14] Final state / validation / failure
experiments/staged_reconstruction/RubberDuck_relative_order/{refresh_repair.md,v1/final_review.json,v1/final_validation.json}

[S15] Training / validation cameras data/derived/kinematic_resplit/{training/transforms_train.json,transforms_val.json}

Current checkpoint SHA-256

J6: 89109bdb3990c95037924fdf0985e073
02ed67065d98971d77209780b96e0502

Current RGB control: 271359bc48db1448c03fa80a5f36a71d
f2379e38461c2bb30fcc8b37ae3c61be

Current order candidate: ccf95ce37ebfd4e5e5ffd163f2d60742
85a5975a5137adf427d703539a093fc7

Both trial checkpoints are in runs/pidg_staged_reconstruction/RubberDuck_relative_order/v1/formal/{rgb_control,ordinal_depth}/checkpoint.pt. Full state is preserved; no automatic resumption or rerun. The J6 parent is runs/pidg_staged_reconstruction/RubberDuck_I10_joint/v1/formal/joint_depth_trust/checkpoint.pt.